

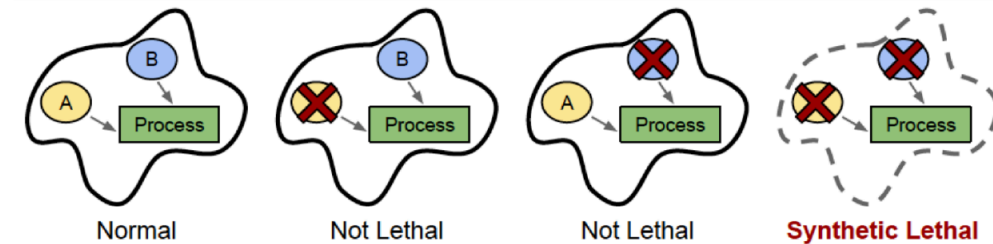
From SL Promise to an Honest Ranking Model

Ranking candidate synthetic-lethal gene pairs in K562

Combining DepMap dependency, Perturb-seq response, and a frozen perturbation foundation model

What is synthetic lethality — and why it's hard

- **SL**: losing either gene alone is tolerated; losing **both** kills the cell
- One gene already inactivated in the tumor → its SL partner becomes a selective drug target
- Clinical proof: **PARP inhibitors in BRCA1/2-mutant tumors**
- The bottleneck: **~200M human gene pairs**, mostly **context-dependent** — can't brute-force experimentally
- → we need a computational way to **rank** candidate partners



The task we actually solve (and what we don't claim)

Benchmark: Feng et al. 2024, SynLethDB-derived · Rand 1:1 balanced · 9,471-gene K562 universe

Split	Held out	Difficulty
CV1	pair-level (genes may recur)	easiest, degree-gameable
CV2	one gene unseen	intermediate
CV3	both genes unseen	cold-start, hardest

- This is **SL-pair classification / ranking**, *not* validated SL target discovery
- Rand negatives are **unconfirmed** non-SL → benchmark adapter, not a K562 SL assay
- **CV2 / CV3** are the only honest generalization surfaces

Data sources — and why each one

Source	Role	Why
DepMap CRISPRGeneEffect (Chronos)	dependency label C	population fitness; K562 = ACH-000551
Replogle K562 gwps Perturb-seq	transcriptomic response	CRISPRi = loss-of-function, modality-aligned with DepMap KO; 1.99M cells, 6,070 genes
ESM2-650M protein embeddings	gene identity	continuous → generalizes to held-out genes (1280-d)
Feng 2024 / SynLethDB	pair label D	the SL-pair benchmark

Why K562: the only cell line with both deep Perturb-seq *and* DepMap dependency.

Stage 1 — observed transcriptome predicts dependency

exp01: pseudobulk Δ -expression $\rightarrow$ PCA + Ridge

- Replogle 5-fold CV $\approx$ **0.49** Spearman · Adamson transfer **0.50** (AUROC 0.886)

exp02 audit: is it just a generic cell-death / viability axis?

- NAR viability score alone **0.244** vs best pseudobulk **0.494**
- NAR-residualized transcriptome still **0.503** $\rightarrow$ signal is **real and transcriptomic**, not just "everything dies"

Stage 2 — learning a dependency-aware representation ← *the bridge*

- **exp03** single-cell set learning: scVI128 + **frozen-GMM distribution regression** wins
 - Adamson Spearman **0.666**, AUROC **0.911** — beats attention MIL; HVG hurts
- **exp04** leakage-free *predicted-B* loop (forward $A \rightarrow B \rightarrow C$, no observed test bag)
- **exp05** frozen-**STATE** forward model (Arc Institute), $A \rightarrow B \rightarrow C$ pipeline

Takeaway: we now have a **STATE-based way to turn any perturbation into a dependency-aware embedding** → carry this into the SL-pair task.

Stage 3 — the dependency-only floor (exp06)

Input: only the two genes' DepMap GeneEffect scalars → 5 swap-invariant features → P(SL)

Model	CV1 NDCG@10	CV2 AUROC	CV2 NDCG@10	CV3 AUROC	CV3 NDCG@10
B (XGBoost)	0.0505	0.704	0.042	0.596	0.002
C (degree probe)	0.197	0.500	0.001	—	—

- Degree probe **wins CV1** → CV1 is gameable by train-positive degree
- CV3 $\approx$ cold-start failure for dependency-only features
- **This is the bar every later model must beat — on CV2/CV3**

Does observed Perturb-seq add lift over dependency-only? (exp07)

Method: augment exp06 (GeneEffect features) with **observed gwps response embeddings** per gene

Coverage crux:

- Replogle K562 gwps: **64% per-gene coverage** (6,070 / 9,471 genes)
- But for *pairs*: **~41% both-covered** under independence; **51.17%** in the actual benchmark rows

Completed Tier 1: PCA-delta mean-pool · zero fallback · **no coverage flag**

Model	CV2 AUROC	CV2 NDCG@10	CV3 AUROC	CV3 NDCG@10
B (exp06 XGB)	0.704	0.042	0.596	0.002
B_transcript	0.751	0.094	0.630	0.001

Read: observed Perturb-seq gives clear **CV2 lift**; CV3 top-k ranking remains unresolved.

e2e DL centerpiece pt.1 — the problem & architecture (exp08)

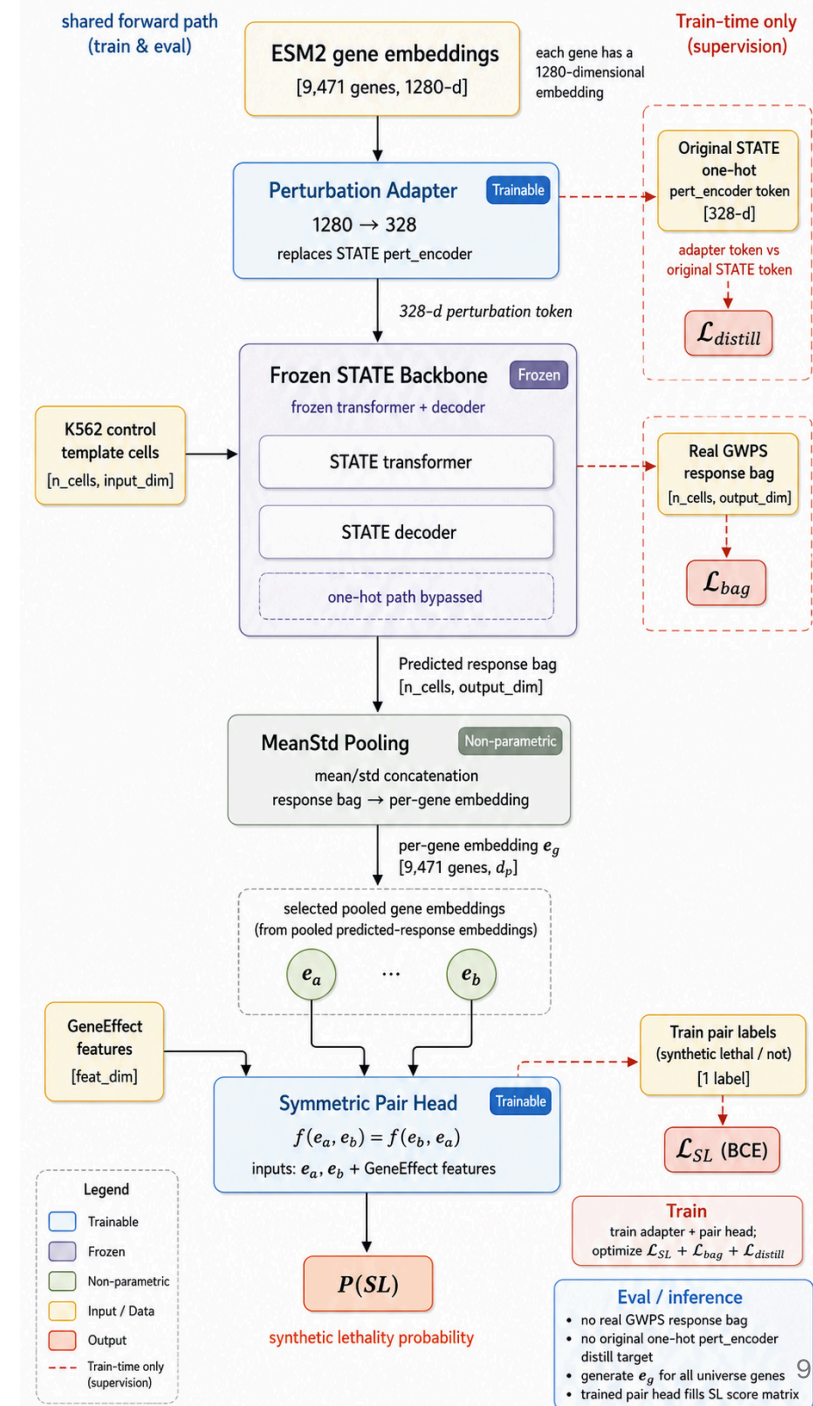
Why exp08: local STATE checkpoint is **closed-vocab one-hot**

- Only **16.3% of the SL universe in-vocab** (1,542 / 9,471 genes)
- 84% out-of-vocab → held-out genes get no gradient

Fix: freeze STATE's 8-layer Llama backbone, **replace** its one-hot `pert_encoder` with a trainable **adapter fed by ESM2** (1280 → 328)

→ all 9,471 genes land in **one coordinate system**

Arch: ESM2 → PertAdapter → frozen STATE → predicted bag → pooling
→ symmetric pair head



e2e DL centerpiece pt.2 — leakage-safe training & the bar

3-part loss:

1. **SL BCE** (pair classification)
2. **Adapter token-distill** (align adapter output to STATE's in-vocab tokens where available)
3. **Real-bag supervision** (covered *train* genes only)

Leakage rule that makes CV2/CV3 valid:

- Held-out genes reached **purely** via `adapter(ESM2)` + frozen STATE
- Never supervised by their own observed response bag → genuinely unseen

The bar to beat: exp06 CV2 AUROC **0.704** / CV3 **0.596**, with lift concentrated on covered-pair slice

Status: code + unit tests complete; **CV2/CV3 cluster gates pending**

A parallel route — cross-cell-line selectivity (exp09)

No transcriptome: use DepMap across **1,208 lines** — is KO(b) more lethal where gene_a is defective?

Composite-OR defective call: mutation / CN loss / low expression

Results:

- CV2 B_xcl: AUROC **0.742** (+0.039 over B), NDCG@10 **0.086** (+0.044)
- CV3 B_xcl: AUROC **0.645** (+0.050), but NDCG@10 **flat** (0.001 vs 0.002)

Read: cross-cell-line selectivity improves cold-start **classification** but does not fix cold-start **ranking**

Non-pan-essential slice: lift shrinks but doesn't vanish on CV1/CV2; CV3 mostly essentiality-linked

Closing the loop — what we built, what we didn't claim

Built: an honest, leakage-controlled SL-pair ranking adapter (classification + ranking), not validated SL target discovery

The recurring discipline:

- Simple floors first (exp06 dependency-only)
- CV2 / CV3 as the real bar (CV1 is degree-gameable)
- Partial results count (exp07 lifts CV2, not CV3 top-k ranking)
- Pending is marked pending (exp08 has no fabricated metrics)

Path to true context-specific SL discovery:

- exp08 cluster results → does frozen-STATE+ESM2 adapter generalize to held-out genes?
- exp09 selectivity → does cross-cell-line evidence add robustness?
- TCGA patient-context transfer → cell line → tumor dependency mapping

Appendix — Results Scoreboard with SOTA Comparison

Model	Source	CV1 F1	CV1 NDCG	CV2 F1	CV2 NDCG	CV3 F1	CV3 NDCG
DDGCN	literature	0.9104	0.2159	0.9113	0.2494	0.9104	0.2470
GRSMF	literature	0.8757	0.5178	0.8905	0.5075	0.8905	0.5075
SL2MF	literature	0.8611	0.2745	0.4332	0.0052	0.4160	0.0001
A (logreg)	exp06	0.6675	0.0040	0.6677	0.0048	0.6686	0.0035
B (XGBoost)	exp06	0.7304	0.0505	0.6756	0.0421	0.6701	0.0024
C (degree probe)	exp06	0.8227	0.1970	0.6667	0.0006	0.6667	0.0008
A_transcript (logreg + Perturb-seq)	exp07	0.6919	0.0268	0.6687	0.0199	0.6674	0.0078
B_transcript (XGB + Perturb-seq)	exp07	0.7695	0.1738	0.7017	0.0942	0.6762	0.0011
A_xcl (logreg + selectivity)	exp09	0.6675	0.0096	0.6676	0.0118	0.6699	0.0081
B_xcl (XGB + selectivity)	exp09	0.7436	0.1601	0.6942	0.0864	0.6727	0.0011
exp08 (frozen-STATE + ESM2 e2e DL)	exp08	pending	pending	pending	pending	pending	pending

Caveat: Literature rows are K562-filtered benchmark references, not a single implementation-harness ablation. Our NDCG = NDCG@10 (official per-anchor protocol). Within-harness lifts: B_transcript (exp07) vs B on CV2 NDCG (0.0942 vs 0.0421); B_xcl (exp09) vs B (0.0864 vs 0.0421).